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|-----------------------------------------------------------|----------------------------------------------------|
| Program: <b>Diploma in Civil Engineering and Planning</b> |                                                    |
| Course Code: <b>4391</b>                                  | Course Title: <b>Fundamentals of Town Planning</b> |
| Semester: <b>4</b>                                        | Credits: <b>4</b>                                  |
| Course Category: <b>Program Core</b>                      |                                                    |
| Periods per week: <b>4 (L:3,T:1,P:0)</b>                  | Periods Per Semester: <b>60</b>                    |

#### Course Objectives:

- To plan the strategies for successful implementation of infrastructure projects in urban area.

#### Course Prerequisites:

| Topic                                            | Course Code | Course name                                      | Semester |
|--------------------------------------------------|-------------|--------------------------------------------------|----------|
| Basic Knowledge of building construction         | 3013        | Building Construction and Construction Materials | S3       |
| Knowledge of Building Drawing and Estimation Lab | 3018        | Building Drawing and Estimation Lab              | S3       |

#### Course Outcomes:

On completion of the course, the student will be able to:

| CO n | Description                                                                                              | Duration (Hours) | Cognitive Level |
|------|----------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1  | To understand the history and basics of town planning, design concepts and site plan preparations        | 16               | Applying        |
| CO2  | To understand the theories of town planning and environmental aspects in planning                        | 14               | Applying        |
| CO3  | To understand the concepts of master plan preparation and various standards of planning                  | 14               | Applying        |
| CO4  | To understand the concepts of replanning, planning of an eco sensitive zone and management for disasters | 14               | Applying        |
|      | Series Test                                                                                              | 2                |                 |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             |     |     |     | 3   |     |     |     |
| CO2             |     |     |     | 3   |     |     |     |
| CO3             |     |     |     | 3   |     |     |     |
| CO4             |     |     |     | 3   |     |     |     |

3-Strongly Mapped, 2-Moderately Mapped,1-Weakly Mapped

**Course Outline:**

On completion of the course,the student will be able to:

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Description                                                                                              | Duration( Hours) | Cognitive Level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>To understand the history and basics of town planning, deisgn concepts and site plan preparations</b> |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Understand the history and development of towns                                                          | 4                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Identify the types of buildings and preparation of site as per requirements                              | 4                | Applying        |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Identify the design requirements for ensuring circulation and ventilation                                | 4                | Understanding   |
| M1.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Identify the preparation of site plans, models and computer aided design and drafting                    | 4                | Applying        |
| <p>History of Town planning: - Principles of town Planning, Town planning in ancient India, Indus valley Civilization, medieval period, necessity of town planning, growth and forms of towns, site selection for an ideal town, urbanization and, industrialization and its effects on town planning.</p> <p>Functional planning of buildings - occupancy classification of buildings - general requirements of site and building codes and rules - licensing of building works - the process of identifying activity areas and linkages – Design concepts and philosophies - checking for circulation, ventilation, structural requirements and other constraints - preparing sketch plans and working drawings - site plans - presentation techniques - pictorial drawings - perspective and rendering - model making - introduction to computer aided design and drafting</p> |                                                                                                          |                  |                 |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>To understand the theories of town planning and environmental aspects in planning</b>                                                    |   |               |
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | identify the theories of town planning and problems of urban growth                                                                         | 4 | Understanding |
| M2.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Understand the Various aspects of town planning Aspects of New town and its comprehensive planning based on surveys and demographic studies | 4 | Understanding |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Identify the environmental aspects and theories of land use planning                                                                        | 3 | Understanding |
| M2.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Familiarize with the urban delineation and urban planning and concepts of regional planning                                                 | 3 | Understanding |
| Town planning theory - evolution of towns - problems of urban growth - beginning of town planning acts - ideal towns - garden city movement - concept of new towns and conservative surgery - comprehensive planning of towns - survey and analysis of town - base maps - land use classification - transportation network – housing, demographic, socio - economic studies - Environmental aspects - theories of land use planning, transportation planning and housing development - urban area delineation - urban influence zone - urban region - concepts of regional planning                                             |                                                                                                                                             |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>To understand the concepts of master plan preparation and various standards of planning</b>                                              |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Familiarize with the preparation of master plan                                                                                             | 4 | Applying      |
| M3.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Identify the various planning standards for traffic, roads and densities                                                                    | 3 | Applying      |
| M3.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Understand the various growth models town planning legislations and municipal acts                                                          | 3 | Applying      |
| M3.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Identify the various urban financing schemes various schemes for slum clearance and pollution control aspects                               | 4 | Applying      |
| Concepts of master plan, structure plan, detailed town planning scheme and action plan, estimating future needs - planning standards for different land use, allocation for commerce, industries, public amenities, open areas etc. - planning standards for density distributions - density zones - planning standards for traffic network - standard of roads and paths - provision for urban growth - growth models - plan implementation - town planning legislation and municipal acts - planning of control development schemes - urban financing - land acquisition - slum clearance schemes - pollution control aspects |                                                                                                                                             |   |               |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>To understand the concepts of replanning , planning of an eco sensitive zone and management for disasters</b> |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Understand the methods of replanning of existing towns and modern trends in town planning                        | 4 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Identify the Planning principles for eco-sensitive zones                                                         | 3 | Understanding |
| M4.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Familiarize the planning & management for disasters                                                              | 3 | Applying      |
| M4.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Understanding about integration of disaster mitigation strategy in Urban and Regional planning                   | 4 | Applying      |
| <p>Replanning of existing towns - planned cities in India - recent trends in town planning</p> <p>Planning principles for eco-sensitive zones (coastal habitat, hill towns, riverfront, etc): Bio-climatic considerations at neighborhood level to city level.</p> <p>Planning &amp; Management for Disasters: Disaster risk, vulnerability and capacity assessment (concepts and methodology), Disaster management and Planning, Hazard safety legislations, role of various stakeholders and community based disaster risk management.</p> <p>Integration of disaster mitigation strategy in Urban and Regional planning.</p> |                                                                                                                  |   |               |

**Text/Reference:**

| <b>T/R</b> | <b>BookTitle/Author</b>                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------|
| T1         | Satish Chandra agarwala, Architecture and Town Planning, Dhanpat Rai & Co                                    |
| T2         | Gurucharan Singh and Jagdish Singh, Building Planning and Scheduling, Standard Publishers and Distributors.. |
| T3         | S.C Rangwala, Town Planning, Charotar Publishing House.                                                      |

**Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                 |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.designingbuildings.co.uk/">https://www.designingbuildings.co.uk/</a>                                                                           |
| 2            | <a href="https://architecture-student.com/architecture/principles-of-town-planning/">https://architecture-student.com/architecture/principles-of-town-planning/</a> |
| 3            | <a href="https://ebooks.inflibnet.ac.in/hsp01/chapter/town-planning-and-townships/">https://ebooks.inflibnet.ac.in/hsp01/chapter/town-planning-and-townships/</a>   |